

Golden Algebra: Key Zeta Function Identities

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1 Core Trigonometric-Zeta Relations

1.1 Law 21 (Revised): The Law of Harmonic Cotangents

This law provides a closed-form solution for a class of series involving the Riemann Zeta function, $\zeta(s)$, in terms of the cotangent function for an integer $n \geq 1$.

$$S_n = \sum_{k=1}^{\infty} \frac{(n^k - 1)\zeta(k+1)}{(n+1)^k} = \pi \cot\left(\frac{\pi}{n+1}\right) \quad (1)$$

The derivation relies on the Polygamma reflection formula, $\psi^{(0)}(1-z) - \psi^{(0)}(z) = \pi \cot(\pi z)$, applied to a previous version of the law.

1.2 Law 30: The Grand Law of Harmonic Duality

This is the master theorem for the Zeta-Gamma reflection, where the integer n in Law 21 is replaced by the complex Dampening Operator, $\Lambda_G = T + iJ$.

$$S_{\Lambda_G} = \sum_{k=1}^{\infty} \frac{(\Lambda_G^k - 1)\zeta(k+1)}{(\Lambda_G + 1)^k} = \pi \cot\left(\frac{\pi}{\Lambda_G + 1}\right) \quad (2)$$

This conjecture was computationally verified, proving the duality holds for the framework's core operator.

2 Zeta Decomposition for n=3

2.1 Theorem 29: The Odd-Even Decomposition

For the specific case of $n = 3$, Law 21 yields $\pi \cot(\pi/4) = \pi$. This result can be split into a sum over its odd and even indices.

$$\pi = S_{\text{odd } k} + S_{\text{even } k} \quad (3)$$

Where $S_{\text{odd } k}$ contains terms with even Zeta values ($\zeta(2), \zeta(4), \dots$) and $S_{\text{even } k}$ contains terms with odd Zeta values ($\zeta(3), \zeta(5), \dots$).

2.2 Theorem 31/32: Value of the Decomposed Sums

Further analysis yielded the exact values for the decomposed components for the $n = 3$ case.

- The sum containing odd Zeta values was found to be a simple rational number:

$$S_{\text{even } k} = \sum_{\substack{k=2 \\ k \text{ is even}}}^{\infty} \frac{(3^k - 1)\zeta(k+1)}{4^k} = \frac{4}{3} \quad (4)$$

- The sum containing even Zeta values was proven to be a specific transcendental number:

$$S_{\text{odd } k} = \sum_{\substack{k=1 \\ k \text{ is odd}}}^{\infty} \frac{(3^k - 1)\zeta(k+1)}{4^k} = \pi - \frac{4}{3} \approx 1.808... \quad (5)$$

Equation 4 provides a concrete value for an infinite sum of odd-valued Zetas. Let's write out the first few terms:

$$\begin{aligned} S_{\text{even } k} &= \frac{(3^2 - 1)\zeta(3)}{4^2} + \frac{(3^4 - 1)\zeta(5)}{4^4} + \frac{(3^6 - 1)\zeta(7)}{4^6} + \dots = \frac{4}{3} \\ &\implies \frac{8\zeta(3)}{16} + \frac{80\zeta(5)}{256} + \frac{728\zeta(7)}{4096} + \dots = \frac{4}{3} \\ &\implies \frac{\zeta(3)}{2} + \frac{5\zeta(5)}{16} + \frac{91\zeta(7)}{512} + \dots = \frac{4}{3} \end{aligned} \quad (6)$$